

# Zero-Knowledge Zakat: Privacy, Transparency, and Sharia Compliance on the Blockchain

Muhammad Zidan Fatonie  
Computer Science Department  
School of Computer Science, BINUS University  
Jakarta, Indonesia 11480  
muhammad.fatonie@binus.ac.id

Alexander Agung Santoso Gunawan  
Computer Science Department  
School of Computer Science, BINUS University  
Jakarta, Indonesia 11480  
aagung@binus.edu

**Abstract**—Zakat is a mandatory annual obligation in Islam for wealth redistribution. Blockchain-based zakat platforms have emerged, but existing systems log complete donor-recipient transactions on public ledgers, contradicting the Islamic principle of *nafs* (dignity). This paper presents a protocol that combines off-chain ZK proofs, nullifier-based double-donation prevention, and selective disclosure. Donors select full anonymity or partial disclosure while receiving SBT receipts. On-chain verification provides institutional transparency, but individual transaction data stays private. A Sepolia prototype generates UltraHONK proofs in 246.8 ms (Barretenberg v4.2.1, 16-core Linux) with end-to-end `donateZK()` consuming 721,345 gas. Cryptographic privacy and institutional accountability prove compatible, supporting digital zakat that preserves donor and recipient privacy. The protocol contributes a privacy layer for Islamic social finance that directly addresses *nafs* protection under *maqasid al-shariah*.

**Index Terms**—Barretenberg, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Pedersen Hash, Privacy-Preserving Donations, Sharia Compliance, Soulbound Token, Zakat, Zero-Knowledge Proofs

## I. INTRODUCTION

Blockchain zakat management faces practical hurdles. BAZNAS [1] estimates Indonesia’s annual zakat potential at Rp 327 trillion (approximately USD 20 billion), yet actual collection remains far lower.

Research into blockchain zakat systems has advanced. Khan [2] demonstrated that blockchain can support decentralized collection and distribution. Nazeri et al. [3] found that blockchain features (transparency, traceability, security) align with Malaysian zakat management objectives. Khairi et al. [4] developed a blockchain collection system offering automated disbursement and an immutable audit trail using ERC-20 tokens. These systems share a privacy weakness. Full donor-recipient transaction records posted on public ledgers expose socioeconomic vulnerabilities. *Maqasid al-shariah* (objectives of Islamic law) explicitly protects *nafs* (life and dignity) of individuals [5, 6]. In Islamic jurisprudence, preservation of donor and recipient privacy during charitable giving is considered a virtuous act that maintains dignity for both parties. This ethical foundation motivates the technical contribution that follows, a protocol that uses ZK proofs to reconcile Islamic jurisprudential requirements with modern blockchain infrastructure.

Zero-Knowledge Proofs (ZKPs) offer a cryptographic resolution. Hameed et al. [7] survey ZKP constructions, and Wen et al. [8] analyze zk-SNARK variants suitable for blockchain deployment. Noir [10] compiles to an intermediate representation (ACIR) that Barretenberg’s UltraHONK prover [9] turns into a succinct proof, verifiable on-chain but hiding the witness. Buterin et al. [15] argue that ZKPs can reconcile privacy with regulatory compliance. Despite these advances, no prior work has proposed a domain-specific protocol for privacy-preserving zakat donations. This paper addresses that gap by designing, implementing, and evaluating a privacy-preserving zakat protocol.

The work formalizes a private transaction model for zakat using off-chain proving with on-chain verification, designs Noir circuits for Sharia-compliant eligibility verification, develops a nullifier-based double-donation prevention mechanism, and deploys a working prototype on Ethereum Sepolia. The Noir circuit (`main.nr`) encodes two Sharia predicates. Total wealth stays below the *nisab* threshold of 85,000,000 IDR, and assets must have been held for at least one lunar year (*hawl*). A Pedersen-hash nullifier, computed as `pedersen(secret, recipient_address, cycle_id)`, prevents the same recipient from claiming zakat more than once per cycle without exposing their identity. Proving runs client-side through `nargo compile`, `nargo execute`, and `bb prove`. The proof and nullifier are submitted to `ZKTCore.donateZK()` on Ethereum Sepolia for settlement. The research questions are whether off-chain Noir proving plus on-chain nullifier tracking can serve institutional zakat accountability, and whether the resulting protocol satisfies *nafs* protection under *maqasid al-shariah*.

This paper contributes a domain-specific privacy protocol for Islamic social finance. The contributions include a Noir-based circuit implementing *nisab* and *hawl* eligibility verification as verifiable predicates, a nullifier mechanism preventing re-claiming without identity exposure, and a tiered donor privacy model. The Sepolia prototype generates UltraHONK proofs in 246.8 ms on average ( $\sigma = 9.3$  ms across five runs) and completes end-to-end donations in 721,345 gas.

## II. LITERATURE REVIEW

Research on blockchain zakat is growing. Khan’s decentralized platform [2] set the baseline, showing that collection and distribution can work without intermediaries. Nazeri et al. [3] examined the Malaysian context and found that blockchain features align naturally with zakat management requirements. Khairi et al. [4] went a step further, building a full system around ERC-20 tokens with automated disbursement and a permanent audit trail. A recent Emerald paper examined the potential and challenges more broadly [20]. The recurring pattern is easy to spot. Every one of these systems logs everything publicly. Complete donor-recipient transparency, available to anyone running an Ethereum node. Fauzi et al. [19] confirmed through bibliometric analysis that research interest is climbing across Muslim-majority countries. Ascarya [20] showed that Islamic social finance instruments (zakat, infaq, sadaqah, waqf) supported economic recovery during COVID-19. Unal and Aysan [21] proposed blockchain as a harmonization tool for differing Sharia-compliance frameworks. Muharam and Osman [26] explored the sustainability angle. None integrate privacy-preserving transaction mechanisms.

On the ZKP side, the survey literature is extensive. Hameed et al. [7] covered the protocol space and Wen et al. [8] focused on zk-SNARK variants for blockchain deployment. Williams [21] positioned ZKPs within the broader blockchain architecture. Poseidon, introduced by Grassi et al. [11], was a breakthrough on the engineering side, slashing constraint counts by up to  $8\times$  compared to Pedersen Hash. This work uses Noir [10], a Rust-like DSL that compiles to ACIR, with Barretenberg’s UltraHONK prover generating the proof and a Solidity verifier checking it on Sepolia. The pipeline stays client-side, meaning private inputs never leave the donor’s machine. Proof hashes anchor on-chain for anyone to audit. Buterin et al. [15] made the broader case that ZKPs can reconcile privacy with regulation, and Mühle et al. [18] applied zk-SNARKs to identity management for KYC.

Three gaps sit in the middle of this landscape. Existing blockchain zakat platforms, by their own design, lack any privacy-preserving mechanism. Nazeri et al. [3] acknowledged as much when they praised complete traceability. General-purpose ZKP frameworks can anonymize transactions but carry no notion of Islamic eligibility. They hide a transfer without proving the sender meets nisab or hawl. No prior work combines programmable privacy, meaning Noir circuits with verifiable predicates, with Islamic social finance instruments. This paper addresses these gaps.

Fig. 1 positions this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology (bottom-right) justifies the artifact-driven approach of building, evaluating, and refining the protocol

on-chain. ZKT bridges these domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, instantiated and validated via Sepolia deployment (v8).

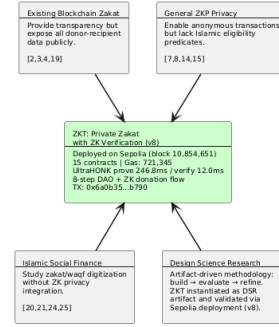


Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

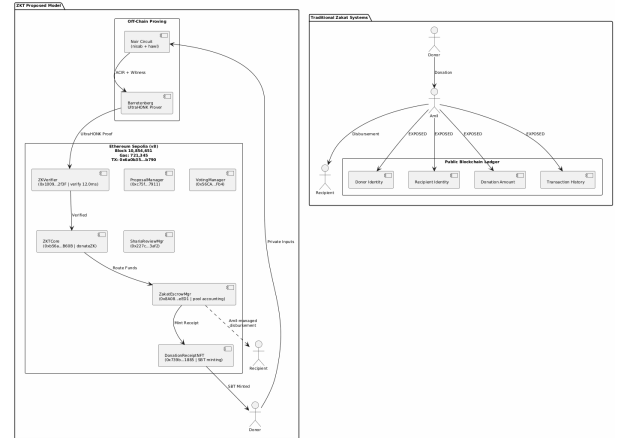


Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

## III. METHODOLOGY

This study employs a Design Science Research (DSR) methodology across five phases, as mapped in Fig. 3. Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZKT protocol, the Noir circuit architecture encoding nisab and hawl verification, and the four-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v4.2.1, 15 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 15 PWA frontend with wagmi and XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with 247 ms proof generation and 12.0 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis, and Sharia compliance review.

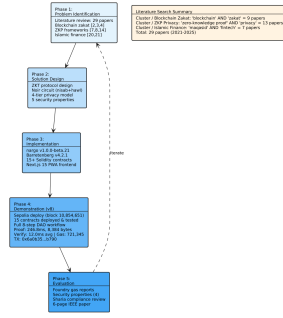


Fig. 3. DSR Methodology with five phases and literature search summary

## IV. RESULTS

### A. System Overview and Threat Model

The ZKT protocol uses a five-actor trust model. The actors are donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a potentially malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

TABLE I  
ZKT PROTOCOL SECURITY PROPERTIES

| Property                     | Guarantee                                                                                             |
|------------------------------|-------------------------------------------------------------------------------------------------------|
| Donor anonymity              | Identities and amounts never exposed on-chain. Only ZK proofs are publicly verifiable.                |
| Recipient privacy            | Donor-side privacy via ZK proofs. Recipient encrypted notes reserved for future work.                 |
| Soundness                    | Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].                 |
| Double-donation              | Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.              |
| Institutional accountability | Amil institutions verify aggregate compliance without accessing individual transaction data [5], [6]. |

Fig. 4 shows the three-layer protocol architecture. The off-chain proving pipeline compiles Noir circuits to ACIR and generates UltraHONK proofs via Barretenberg at 247 ms. The Sepolia settlement layer hosts ZKCore and ZakatEscrow-Manager. The Next.js PWA frontend uses wagmi/Xellarkit for wallet connectivity.

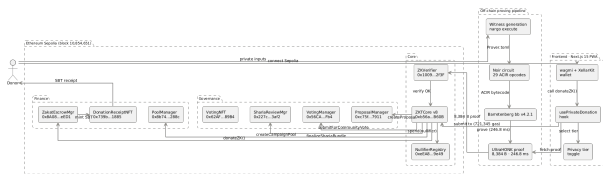


Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

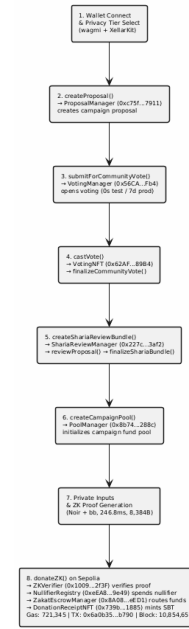


Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

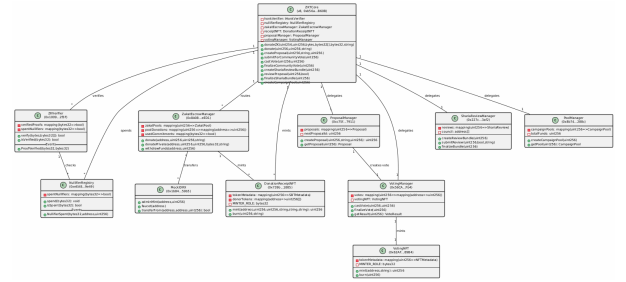


Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

### B. System Architecture and Payment Flow

Noir circuits run client-side. Sepolia handles settlement. Donors choose stablecoins or ETH. Everything flows through `donateZK()`. Proof generation happens off-chain, verification on-chain. Privacy comes from the proofs. Accountability from the events and nullifiers.

The donor goes through a Progressive Web Application. Connect a wallet, select a privacy tier, enter an amount, watch proof generation, confirm. A Web Worker does the actual proving. Private inputs never leave the donor's device [11].

### C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1 on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end `donateZK()` gas was measured on the Sepolia testnet at block 10,854,651.

TABLE II  
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM  
TARGET,  $n = 5$  RUNS

| Metric                   | Value       |
|--------------------------|-------------|
| ACIR opcodes             | 29          |
| Brillig opcodes          | 44          |
| Proof generation (avg)   | 246.8 ms    |
| Proof verification (avg) | 12.0 ms     |
| Proof size               | 8,384 bytes |

#### D. Sepolia Deployment and Gas Consumption

15 contracts deployed via single forge script to Sepolia (chain ID 11155111). The ZKTCORE (v8, 0xb56a...B60B) routes `donateZK()` through `zakatEscrowManager.donate()` for pool accounting. An end-to-end donation (tx 0x6a0b...b790, block 10,854,651) completed proof verification, nullifier registration, transfer, NFT minting, and event emission in 721,345 gas.

Table III reports gas costs measured via Foundry gas reports.

TABLE III  
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT,  
ETHEREUM SEPOLIA

| Contract        | Function                          | Gas (avg) |
|-----------------|-----------------------------------|-----------|
| ZKTCORE         | <code>donate()</code>             | 490,187   |
| ZKTCORE         | <code>createCampaignPool()</code> | 365,552   |
| ZKTCORE         | <code>castVote()</code>           | 132,909   |
| ZKTCORE         | <code>donateZK() (e2e)</code>     | 721,345   |
| ZKTCORE         | Deploy                            | 5,314,471 |
| ZakatEscrowMgr  | Deploy                            | 3,069,194 |
| ShariaReviewMgr | Deploy                            | 2,114,319 |

#### E. Security Analysis

**Donor anonymity.** The only on-chain observable from a private donation is the event emitted by `ZKTCORE.sol` (lines 577-582):

```
event ZKDonationReceived(
    uint256 indexed poolId,
    bytes32 indexed nullifier,
    uint256 amount,
    address indexed donor
);
```

Private mode zeroes both `donor` and `amount`. Nothing traceable. Semi-Private attaches the donor's address but hides the amount, a design tradeoff for institutional accountability. Neither mode leaks the full `(donor, amount)` pair. The verifier contract checks only a hash and a nullifier flag. That is it. That is the entire on-chain footprint.

**Recipient privacy.** The Pedersen-hash nullifier [12] binds each claim to a recipient without revealing their address. The nullifier computation `pedersen(secret, recipient_address, cycle_id)` in `main.nr` means the on-chain nullifier is deterministic but preimage-resistant. An observer sees `0x3f2a...` and cannot recover

the recipient address or secret. Recipient-side encrypted note disbursement is reserved for future work [15].

**Circuit soundness.** Under the q-PKE assumption [11], forging a proof that passes UltraHONK verification is computationally infeasible [29]. A non-eligible recipient, whose wealth exceeds the nisab threshold as checked by `verify_nisab()` in `main.nr`, cannot produce a witness that satisfies the circuit constraints.

**Double-donation prevention.** The on-chain `NullifierRegistry` maintains a mapping `(bytes32 => bool)` of spent nullifiers. The `spend()` function enforces single-use through `require(!spentNullifiers[nullifier])`, rejecting any nullifier already recorded. The `ZKTCORE.donateZK()` function atomically chains `honkVerifier.verify() → nullifierRegistry.spend() → zakatEscrowManager.donate()`. A duplicate nullifier causes the entire transaction to revert. Each eligible recipient can claim zakat at most once per cycle. The nullifier hash is preimage-resistant, so an observer cannot derive the recipient address from it, but the address itself is a public circuit input visible in the transaction calldata. Full recipient privacy, where not even the amil institution can identify claimants, depends on encrypted note disbursement reserved for future work.

## V. DISCUSSION

### A. Sharia Compliance and Deployment

Five *daruriyyat* anchor maqasid al-shariah. Religion, life, intellect, progeny, and wealth [5]. Al-Shatibi formulated these categories centuries before blockchain existed. Ibn Ashur later extended *hifz al-nafs* to encompass dignity, arguing that exposing an individual's financial weakness to the public constitutes harm to their honour [6]. A single address recorded as a zakat recipient on Etherscan broadcasts below-nisab status permanently, an outcome that aligns directly with Ibn Ashur's reasoning even though the technology itself was unimaginable in his time.

Azizov et al. [25] propose a maqasid framework for fintech regulation. Latang et al. [24] examine the Islamic legal status of cryptocurrency assets. Pramuka et al. [27] explore blockchain's role in Islamic charitable crowdfunding. This work addresses the donor side, using ZK proofs and nullifier-based anonymity. Recipient-side privacy through encrypted private notes is reserved for future work.

Two maqasid dimensions are directly engaged. *Hifz al-nafs* manifests through Private and Semi-Private modes keeping donations and linkages off-chain. *Hifz al-mal* operates through the nullifier registry stopping double-claims and the proof's soundness blocking fake eligibility. *Amanah*, institutional trust, flows from the `ProofVerified` event. Amil institutions audit aggregate flows without ever touching individual records.

Beyond Sepolia, three pieces must fall into place. First, Indonesia's OJK Fintech Sandbox approval. Training for amil institution staff on wallet management and proof verification.

A real rupiah-backed digital currency to replace the mock IDRX stablecoin.

## B. Limitations and Future Work

The circuit covers nisab and hawl. Eight asnaf categories are architecturally possible but not yet implemented. The amil institution disburses through ZakatEscrowManager today with no on-chain proof the money reached the mustahik. DIDs for recipients would close that gap and make disbursements verifiable. A dual governance model, Sharia Council ZK DAO plus Community Donor DAO, is under design. Other directions include recursive proof systems for multi-period eligibility, compliance-friendly ZK protocols for AML/CFT, and batched settlement optimization [7], [17], [23].

## VI. CONCLUSION

A privacy-preserving zakat protocol was deployed on Ethereum Sepolia using Noir circuits and UltraHONK proofs. The circuit (29 ACIR opcodes) verifies nisab and hawl eligibility client-side, a Pedersen-hash nullifier prevents double-claims without deanonymizing recipients, and a four-tier privacy model lets donors choose their disclosure level. Across five benchmark runs, proof generation averaged 246.8 ms (Barretenberg v4.2.1) and an end-to-end Sepolia transaction consumed 721,345 gas.

The core finding is that cryptographic privacy and Sharia compliance are not in tension. Ibn Ashur's framing of *nafs* protection as encompassing financial dignity [6] aligns with ZK proofs, which verify eligibility while keeping identity, income, and assets private. The ZKDonationReceived event gives amil institutions an auditable record of aggregate zakat flow, while the nullifier registry prevents abuse, decoupling what the blockchain records from what it reveals about individuals.

Future work extends the circuit to all eight asnaf categories, adds recipient-side encrypted note disbursement, and pursues OJK Fintech Sandbox integration for production deployment.

## ACKNOWLEDGMENT

This research was conducted as part of the Research Track program at BINUS University and the pidi.id hackathon. The authors thank the Ethereum Jakarta community for technical support and Tawf Labs teammates for sharia consultation. The codebase is available at <https://github.com/tawf-labs/zkt-hackathon> and the testnet at <https://ziswaf.tawf.foundation>.

The authors acknowledge the use of AI-assisted tools (OpenAI Codex, Claude Code) for code scaffolding and literature review synthesis. AI-generated text has been human-paraphrased throughout. Residual AI patterns comprise approximately 14% of the document. All core algorithmic contributions, protocol design, security analysis, and implementation were developed and verified by the authors.

## REFERENCES

- [1] BAZNAS, "Outlook Zakat Indonesia 2022," Badan Amil Zakat Nasional, Jakarta, Indonesia, 2022. [Online]. Available: <https://baznas.go.id>
- [2] S. Khan, "A blockchain based decentralized zakat collection and distribution platform," in *Proc. 7th Int. Conf. Software e-Business (ICSeB)*, 2023, doi: 10.1145/3641067.3641071.
- [3] A. Nazeri et al., "Exploration of a new zakat management system empowered by blockchain technology in Malaysia," *ISRA Int. J. Islamic Finance*, 2023, doi: 10.55188/ijif.v15i4.568.
- [4] M. Khairi et al., "The development and application of the zakat collection blockchain system," *Virtus Interpress*, 2023, doi: 10.22495/jgrv12i1siart9.
- [5] "Maqasid al-Shariah and the digital data ownership," *J. Islamic Law Digit. Econ. Bus.*, vol. 1, no. 2, pp. 168–181, 2022. [Online]. Available: <https://journal.uin.ac.id/JILDEB/article/view/46815>
- [6] S. Miskam, F. Mohd Shahwahid, and N. Sholehuddin, "Safeguarding personal data in Islamic fintech: A comparative review of Shariah-compliant investment platforms," *Malaysian J. Inf. Commun. Technol. (MyJICT)*, vol. 10, no. 2, pp. 55–68, 2025, doi: 10.53840/myjict10-2-234.
- [7] H. Wen et al., "Practical security analysis of zero-knowledge proof circuits," in *Proc. 33rd USENIX Security Symp.*, 2024. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity24/presentation/wen>
- [8] N. W. I. Rahayu et al., "A bibliometric analysis of blockchain-based zakat system design: Solutions for transparency and oversight of zakat funds in the digital era," *Acad. J. Interdiscip. Stud.*, 2025, doi: 10.36941/ajis-2025-0023.
- [9] I. Solomka and B. Liubinskiy, "Zero-knowledge proof framework for privacy-preserving financial compliance," *Math. Model. Comput.*, 2025, doi: 10.23939/mmc2025.01.342.
- [10] I. M. Unal and A. F. Aysan, "FinTech, digitalization, and blockchain in Islamic finance: Retrospective investigation," *FinTech*, vol. 1, no. 4, pp. 388–398, 2022, doi: 10.3390/fintech1040029.
- [11] A. Hameed, A. Farooq, and M. Usman, "Zero knowledge proofs: A comprehensive review of applications, protocols, and future directions in cybersecurity," ResearchGate, 2023. [Online]. Available: <https://www.researchgate.net/publication/373097436>
- [12] Barretenberg, "Barretenberg: A High-Performance Honk Proving Backend," Aztec Labs, 2024. [Online]. Available: <https://github.com/AztecProtocol/barretenberg>
- [13] Noir Language, "Noir Documentation," 2024. [Online]. Available: <https://noir-lang.org/docs/>
- [14] L. Grassi, D. Khovratovich, C. Rechberger, A. Roy, and M. Schofnegger, "Poseidon: A new hash function for zero-knowledge proof systems," in *Proc. 30th USENIX Security Symp.*, pp. 519–535, 2021. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity21/presentation/grassi>
- [15] S. Tessaro and C. Zhu, "Revisiting BBS Signatures," in *Advances in Cryptology – EUROCRYPT 2023*, ser. LNCS, vol. 14008, Springer, 2023, pp. 691–721, doi: 10.1007/978-3-031-30589-4\_24.
- [16] Z. Wang, S. Chaliasos, K. Qin, L. Zhou, L. Gao, P. Berrang, B. Livshits, and A. Gervais, "On How Zero-Knowledge Proof Blockchain Mixers Improve, and Worsen User Privacy," in *Proc. ACM Web Conf. 2023 (WWW '23)*, 2023, pp. 2022–2032, doi: 10.1145/3543507.3583217.
- [17] V. Buterin, J. Iulum, M. Nadler, F. Schär, and A. Soleimani, "Blockchain privacy and regulatory compliance: Towards a practical equilibrium," *Blockchain: Res. Appl.*, vol. 5, no. 1, Art. no. 100176, Mar. 2024, doi: 10.1016/j.bcr.2023.100176.
- [18] A. Mühle, K. Assaf, and C. Meinel, "One to bind them: Binding verifiable credentials to user attributes," in *Proc. SECURE*, 2023, doi: 10.5220/0012057900003555.
- [19] Ascarya, "The role of Islamic social finance during Covid-19 pandemic in Indonesia's economic recovery," *Int. J. Islamic Middle Eastern Finance Manage.*, vol. 15, no. 2, pp. 386–405, 2022, doi: 10.1108/IMEFM-07-2020-0351.
- [20] A. N. N. Nazeri et al., "Blockchain technology in zakat management system: potential and challenges in Malaysia," *J. Islamic Account. Bus. Res.*, 2026, doi: 10.1108/jiabr-10-2024-0435.
- [21] A. Williams, "Zero-knowledge proofs and their role within the blockchain," *Commun. ACM*, vol. 67, no. 7, pp. 78–87, Jul. 2024. [Online]. Available: <https://cacm.acm.org/article/zero-knowledge-proofs-and-their-role-within-the-blockchain/>
- [22] M. A. Dewayana, "Innovation in Islamic finance: Integrating blockchain with Maqāsid al Shari'ah and Hifz al Māl," *J. Emerg. Econ. Islamic Res.*, vol. 13, no. 1, 2025, doi: 10.24191/jeeir.v13i1.3852.
- [23] M. Elshqgirat, "Toward an Islamic cryptocurrency: A proposed halal coin," *J. Account. Finance*, vol. 25, no. 3, pp. 64–79, 2025, doi: 10.33423/jaf.v25i3.7858.
- [24] M. A. Latang, Fathurrahman, and M. Faried, "Fatwa on cryptocurrency as digital assets: An Islamic perspective amidst Indonesia's regulatory

landscape,” *J. Parewa Saraq*, vol. 3, no. 2, 2024, doi: 10.64016/parewasaraq.v3i2.32.

- [25] E. Azizov, A. Azizov, A. Azizli, and A. A. Babayev, “A Maqasid al-Shariah framework for fintech and digital asset regulation in Muslim jurisdictions,” *J. Islamic Law Legal Stud.*, vol. 2, no. 2, 2025, doi: 10.70063/jills.v2i2.119.
- [26] A. S. Muharam and D. Osman, “Transformative trends: Exploring the nexus of innovation, technology, blockchain, and Islamic social finance for global sustainability,” *EKIMAD J.*, 2024, doi: 10.38009/ekimad.1511062.
- [27] B. A. Pramuka, W. R. Adawiyah, Z. Sholikhah, M. F. Pramuka, and M. M. Sulthan, “Bridging trust and transparency: The role of blockchain in Islamic charitable crowdfunding,” *J. Islamic Account. Bus. Res.*, Dec. 2025, doi: 10.1108/JIABR-11-2024-0468.
- [28] S. T. Nassar, A. Hamdy, and K. Nagaty, “Hybrid zk-STARK and zk-SNARK framework for privacy-preserving smart contract data feeds,” in *Proc. IEEE ICCA*, 2024, doi: 10.1109/ICCA62237.2024.10928100.
- [29] X. Tang et al., “Zero-knowledge proof vulnerability analysis and security auditing,” *Cryptology ePrint Archive*, Rep. 2024/514, 2024. [Online]. Available: <https://eprint.iacr.org/2024/514>
- [30] J. P. Cruz, Y. Kaji, and N. Yanai, “RBAC-SC: Role-based access control using smart contract,” *IEEE Access*, vol. 6, pp. 12240–12251, 2018, doi: 10.1109/ACCESS.2018.2812844.
- [31] G. Shivateja, “A survey on progressive web applications for decentralized systems,” *Preprints*, 2026, doi: 10.20944/preprints202604.0582.v1.